Practical 8: KNN Classification & Evaluation - Submission Template

Practical: 8 of 15 | **Status:** Complete

Student Info

Field	Value
Name	Shravya Pravin Mushan
Enrollment/Roll No.	3264
Date	01/03/2026
Batch/Division	B

Practical Objective

Implement K-Nearest Neighbors (KNN) algorithm for classification, evaluate its performance with different K values, compare distance metrics, and analyze results through visualizations.

Tests: 5/5 PASSED ✅

Data loaded and explored successfully
Features normalized/scaled correctly
KNN trained with K=3, 5, 7, 9
Distance metrics tested (Euclidean, Manhattan)
Cross-validation completed (5-fold)
Evaluation metrics calculated
Visualizations generated
Analysis documented

Results Summary

Dataset Information

Metric	Value
Dataset	Iris
Total Samples	150
Training Samples (80%)	120
Testing Samples (20%)	30
Number of Features	4
Number of Classes	3

Model Performance

Accuracy Results

K Value	Euclidean	Manhattan	Minkowski
K=3	93.33%	93.33%	93.33%
K=5	93.33%	93.33%	93.33%
K=7	93.33%	93.33%	93.33%
K=9	93.33%	93.33%	93.33%

Cross-Validation Results (5-Fold)

K Value	Mean CV Score	Std Deviation	Best Score	Worst Score
K=3	0.9533	~0.02	~0.98	~0.92
K=5	0.9600	~0.02	~0.99	~0.93
K=7	0.9533	~0.02	~0.98	~0.92
K=9	0.9600	~0.02	~0.99	~0.93

Best Model Performance

Metric	Value
Best K Value	K= 5
Best Distance Metric	Euclidean
Best Test Accuracy	93.33%
Best CV Score	0.9600
Precision (Macro)	0.93
Recall (Macro)	0.93
F1-Score (Macro)	0.93

Detailed Evaluation Metrics (Best Model)

Confusion Matrix

	Predicted		
	Class 0	Class 1	Class 2
Actual Class 0	10	0	0
Class 1	0	9	1
Class 2	0	1	9

Per-Class Metrics

Class	Precision	Recall	F1-Score	Support
Setosa	1.00	1.00	1.00	10
Versicolor	0.90	0.90	0.90	10
Virginica	0.90	0.90	0.90	10
Weighted Avg	0.93	0.93	0.93	30

Observations & Analysis

K Value Analysis

What impact did different K values have on accuracy and cross-validation scores?

[Your observations here]

- K=3: High accuracy but slightly unstable (risk of overfitting).
- K=5: Best balance of bias-variance, highest CV score.
- K=7: Slight drop in CV accuracy (more smoothing).
- K=9: Similar to K=5 but no clear improvement.

Distance Metric Comparison

How did different distance metrics (Euclidean, Manhattan, Minkowski) perform?

Metric	**Performance**	**Observation**
Euclidean	93.33%	Best and stable for scaled features
Manhattan	93.33%	Similar performance
Minkowski	93.33%	Same as Euclidean for p=2

Optimal K Value Selection

Why did you choose **K=5** as the optimal value?

- Training Accuracy: ~96%
- Test Accuracy: 93.33%
- CV Score: 0.9600
- Reasoning: Best cross-validation score with stable test accuracy → good generalization.

Overfitting/Underfitting Analysis

- Is your model overfitting? (Compare train vs test accuracy) No - train vs test gap is small
- Is your model underfitting? (Low overall accuracy) No - accuracy is high

- Cross-validation consistency: Good - scores stable across folds

---

Visualizations Included

Accuracy vs K Value plot
Distance metric comparison chart
Cross-validation scores plot
Confusion matrix heatmap
ROC curves (multi-class)
Feature importance/distance distribution
Decision boundary visualization (2D projections)

Advantages & Limitations

Advantages of KNN

1. Simple and easy to understand
2. No training time (lazy learner)
3. Works very well on small datasets

Limitations of KNN

1. Slow prediction for large datasets
2. Sensitive to feature scaling
3. Performance drops in high dimensions

Hyperparameter Tuning Summary

Parameters Tested:

- K values: 3, 5, 7, 9
- Distance metrics: Euclidean, Manhattan, Minkowski
- Cross-validation folds: 5
- Train-test split: 80-20

Best Configuration:

KNeighborsClassifier(

```
    n_neighbors=5,
    metric='euclidean',
    weights='uniform'
)
```

Code Implementation Summary

Main Libraries Used

pandas, numpy, matplotlib, seaborn, scikit-learn

Key Steps Implemented

1. Loaded dataset
2. Applied StandardScaler
3. Performed stratified train-test split
4. Tested multiple K values
5. Used cross-validation
6. Compared distance metrics
7. Plotted accuracy vs K

Learning Outcomes Achieved

Understand KNN algorithm principles
Implement KNN classifier successfully
Evaluate using appropriate metrics
Compare different K values
Analyze distance metrics
Interpret evaluation results
Document findings clearly

Challenges Faced & Solutions

Challenge	**Solution**
Features on different scales	Applied StandardScaler

Challenge	Solution
Choosing best K	Used cross-validation
Metric confusion	Compared Euclidean, Manhattan, Minkowski

Future Improvements

1. Try weighted KNN
2. Test different distance metrics
3. Apply KNN to a larger dataset

References Used

1. Cover & Hart (1967) Nearest Neighbor Pattern Classification
2. Scikit-learn Documentation
3. Andrew Ng – Machine Learning

Self-Assessment

Aspect	Rating	Comment
Code Quality	⭐⭐⭐⭐⭐	Clean and modular
Documentation	⭐⭐⭐⭐⭐	Well explained
Results Analysis	⭐⭐⭐⭐⭐	Metrics + CV used
Effort & Completeness	⭐⭐⭐⭐⭐	All tasks completed

Submitted By

Item	Details
Student Name	Shravya Pravin Mushan

Item	Details
Signature	
Date	01/03/2026
Instructor Name	Aakash Chatake
Verified	☐

Status: ✅ Complete

Date Submitted: 01/03/2026
Submission Version: 1.0